

# Sequential Decision

## Session 2: Upper confidence bounds

---

Igor Colin

October 2026

ESIEE Paris

# Session 1

---

1. Two arms have means 0.7 and 0.5. Over  $T = 1000$  rounds, an agent pulls the second arm 100 times on average. Compute its regret.
2. Two Bernoulli arms have means 0.6 and 0.4. Give the probability that greedy never plays the first arm again after round 1.
3. ETC with  $m = m^*$  has a regret of order  $\log(T)/\Delta$ . Name the two quantities needed to compute  $m^*$ .

1. Two arms have means 0.7 and 0.5. Over  $T = 1000$  rounds, an agent pulls the second arm 100 times on average. Compute its regret.

$$R_T = \Delta_2 \mathbb{E}[N_2(T)] = 0.2 \times 100 = 20.$$

2. Two Bernoulli arms have means 0.6 and 0.4. Give the probability that greedy never plays the first arm again after round 1.
3. ETC with  $m = m^*$  has a regret of order  $\log(T)/\Delta$ . Name the two quantities needed to compute  $m^*$ .

1. Two arms have means 0.7 and 0.5. Over  $T = 1000$  rounds, an agent pulls the second arm 100 times on average. Compute its regret.

$$R_T = \Delta_2 \mathbb{E}[N_2(T)] = 0.2 \times 100 = 20.$$

2. Two Bernoulli arms have means 0.6 and 0.4. Give the probability that greedy never plays the first arm again after round 1.

$$\mathbb{P}(X_1 = 0, X_2 = 1) = 0.4 \times 0.4 = 0.16.$$

3. ETC with  $m = m^*$  has a regret of order  $\log(T)/\Delta$ . Name the two quantities needed to compute  $m^*$ .

1. Two arms have means 0.7 and 0.5. Over  $T = 1000$  rounds, an agent pulls the second arm 100 times on average. Compute its regret.

$$R_T = \Delta_2 \mathbb{E}[N_2(T)] = 0.2 \times 100 = 20.$$

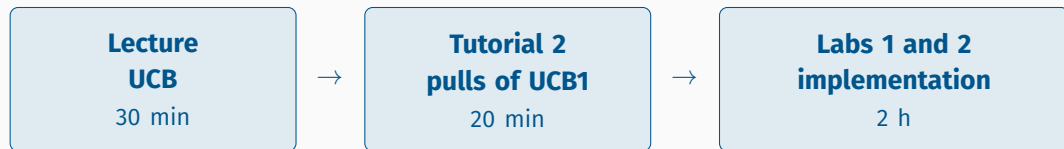
2. Two Bernoulli arms have means 0.6 and 0.4. Give the probability that greedy never plays the first arm again after round 1.

$$\mathbb{P}(X_1 = 0, X_2 = 1) = 0.4 \times 0.4 = 0.16.$$

3. ETC with  $m = m^*$  has a regret of order  $\log(T)/\Delta$ . Name the two quantities needed to compute  $m^*$ .

The horizon  $T$  and the gap  $\Delta$ , which is unknown in practice.

# Outline of the session



- A single notebook covers Labs 1 and 2 (`lab01-02.ipynb`)
- seqlearn framework, greedy,  $\epsilon$ -greedy, ETC, then UCB and the comparison of all the strategies
- Lab 0 (optional) covers installation, numpy, matplotlib and classes

# Strategies of session 1

| Strategy                                   | Needs to know    | Regret              |
|--------------------------------------------|------------------|---------------------|
| Uniform exploration                        | nothing          | linear              |
| Greedy                                     | nothing          | linear (locking)    |
| $\varepsilon$ -greedy, fixed $\varepsilon$ | nothing          | linear              |
| ETC, $m \approx T^{2/3}$                   | $T$              | $O(T^{2/3})$        |
| ETC, $m = m^*$                             | $T$ and $\Delta$ | $O(\log(T)/\Delta)$ |

## Exploration of these strategies

- Decided in advance ( $m$  rounds per arm) or at random (probability  $\varepsilon$ )
- The same for every arm, whatever has been observed

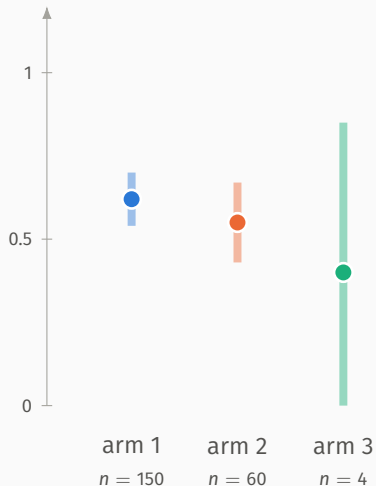
## This session

- Exploration driven by the **uncertainty** on each arm
- Regret in  $\log T$  without knowing  $\Delta$  nor  $T$

# Optimism

---

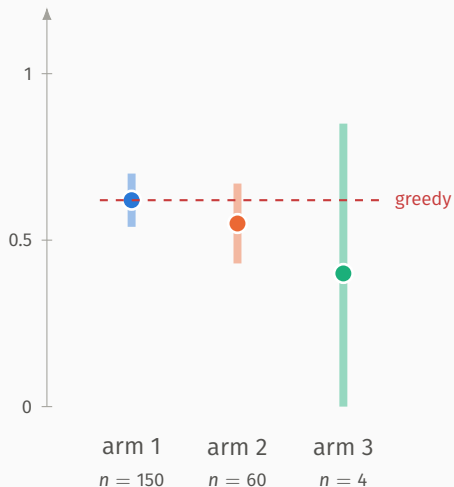
# Confidence intervals



After  $n$  pulls of an arm with rewards in  $[0, 1]$ , Hoeffding gives, with probability at least  $1 - \delta$ ,

$$\mu_a \leq \hat{\mu}_a + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

# Confidence intervals

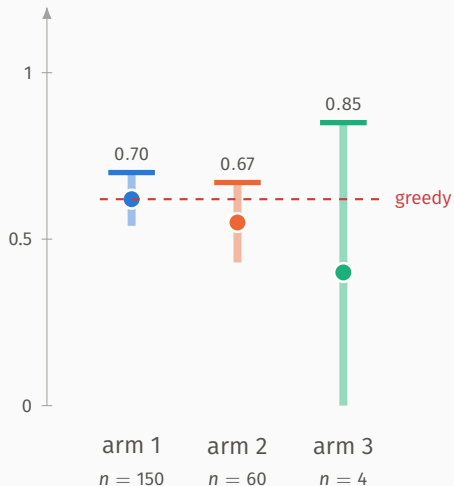


After  $n$  pulls of an arm with rewards in  $[0, 1]$ , Hoeffding gives, with probability at least  $1 - \delta$ ,

$$\mu_a \leq \hat{\mu}_a + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

- Greedy only compares the empirical means, and ignores that arm 3 is almost unknown

# Confidence intervals



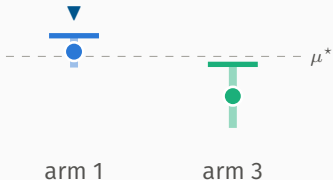
After  $n$  pulls of an arm with rewards in  $[0, 1]$ , Hoeffding gives, with probability at least  $1 - \delta$ ,

$$\mu_a \leq \hat{\mu}_a + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

- Greedy only compares the empirical means, and ignores that arm 3 is almost unknown
- The upper end of the interval is the largest value of  $\mu_a$  that the data do not rule out: an **upper confidence bound**

# The optimism principle

**Principle.** Act as if each arm were as good as the data allow: play the arm with the largest upper confidence bound.

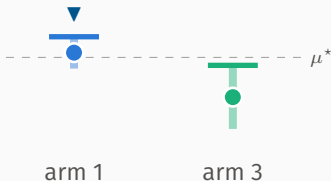


## Case 1: the chosen arm is the best one

- No regret at this round

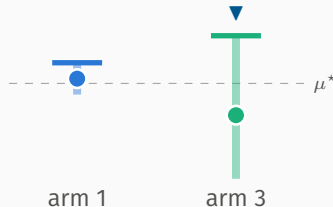
# The optimism principle

**Principle.** Act as if each arm were as good as the data allow: play the arm with the largest upper confidence bound.



## Case 1: the chosen arm is the best one

- No regret at this round

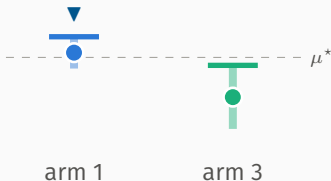


## Case 2: the chosen arm is suboptimal

- Its bound is large because it has been pulled **rarely**

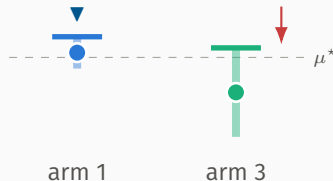
# The optimism principle

**Principle.** Act as if each arm were as good as the data allow: play the arm with the largest upper confidence bound.



## Case 1: the chosen arm is the best one

- No regret at this round



## Case 2: the chosen arm is suboptimal

- Its bound is large because it has been pulled **rarely**
- The pull costs  $\Delta_a$ , but the interval shrinks

**Strategy (Auer, Cesa-Bianchi and Fischer, 2002)**

Pull each arm once. Then, at each round  $t > K$ , play

$$A_t \in \arg \max_a U_a(t) \ , \quad U_a(t) = \hat{\mu}_a(t-1) + \sqrt{\frac{2 \log t}{N_a(t-1)}} \ .$$

**Strategy (Auer, Cesa-Bianchi and Fischer, 2002)**

Pull each arm once. Then, at each round  $t > K$ , play

$$A_t \in \arg \max_a U_a(t) \ , \quad U_a(t) = \hat{\mu}_a(t-1) + \sqrt{\frac{2 \log t}{N_a(t-1)}} \ .$$

**Empirical mean**

- Exploitation: arms that paid well so far

**Strategy (Auer, Cesa-Bianchi and Fischer, 2002)**

Pull each arm once. Then, at each round  $t > K$ , play

$$A_t \in \arg \max_a U_a(t) \ , \quad U_a(t) = \hat{\mu}_a(t-1) + \sqrt{\frac{2 \log t}{N_a(t-1)}} \ .$$

**Empirical mean**

- Exploitation: arms that paid well so far

**Exploration bonus**

- Decreases when the arm is pulled
- Grows slowly while it is left aside

**Strategy (Auer, Cesa-Bianchi and Fischer, 2002)**

Pull each arm once. Then, at each round  $t > K$ , play

$$A_t \in \arg \max_a U_a(t) \ , \quad U_a(t) = \hat{\mu}_a(t-1) + \sqrt{\frac{2 \log t}{N_a(t-1)}} \ .$$

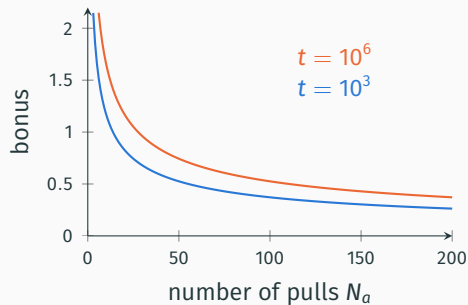
**Empirical mean**

- Exploitation: arms that paid well so far
- The bonus is the Hoeffding half-width with  $\delta = t^{-4}$
- Neither  $\Delta$  nor  $T$  appears in the index
- UCB( $\alpha$ ): bonus  $\sqrt{\alpha \log(t)/N_a(t-1)}$ , and UCB1 is  $\alpha = 2$

**Exploration bonus**

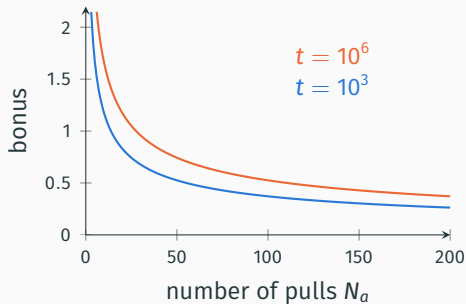
- Decreases when the arm is pulled
- Grows slowly while it is left aside

# The exploration bonus

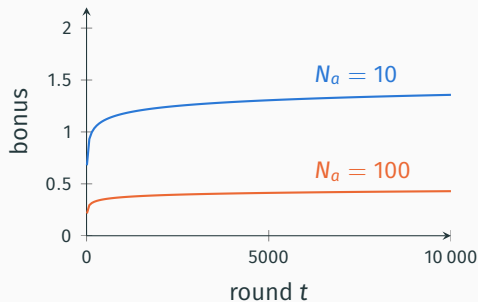


Fixed round  $t$ : decrease in  $1/\sqrt{N_a}$

# The exploration bonus



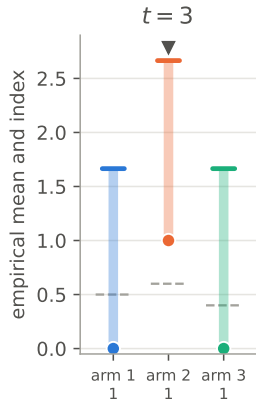
Fixed round  $t$ : decrease in  $1/\sqrt{N_a}$



Fixed number of pulls: slow growth in  $\sqrt{\log t}$

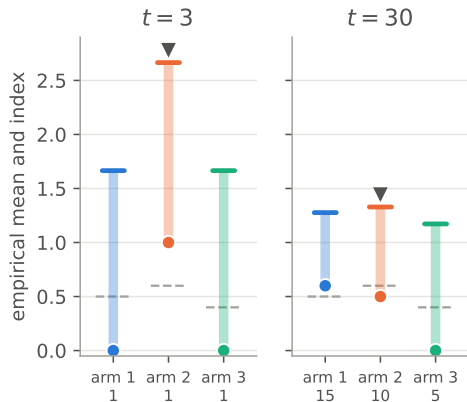
- An arm that is not pulled sees its index grow, so every arm is pulled again eventually: no locking

# One run of UCB1



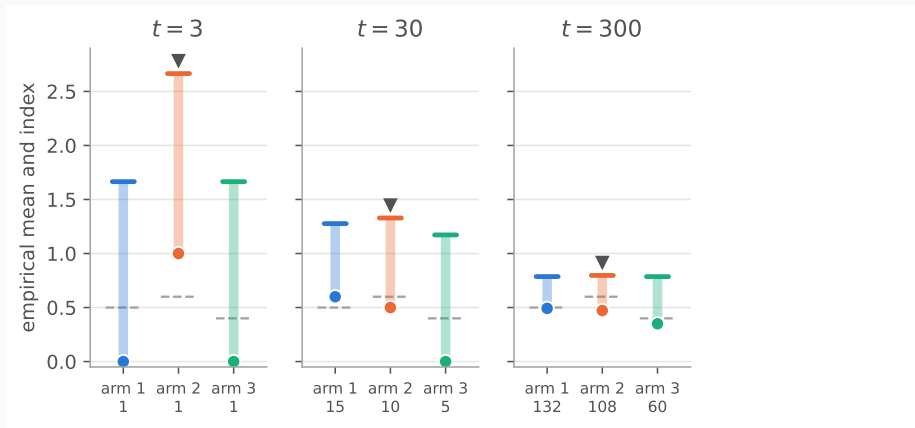
Three Bernoulli arms with means (0.5, 0.6, 0.4) (dashed), after  $t$  rounds. Dots: empirical means; bars: indices  $U_a(t)$ ; ▼: arm played at the next round; below each arm: number of pulls.

# One run of UCB1



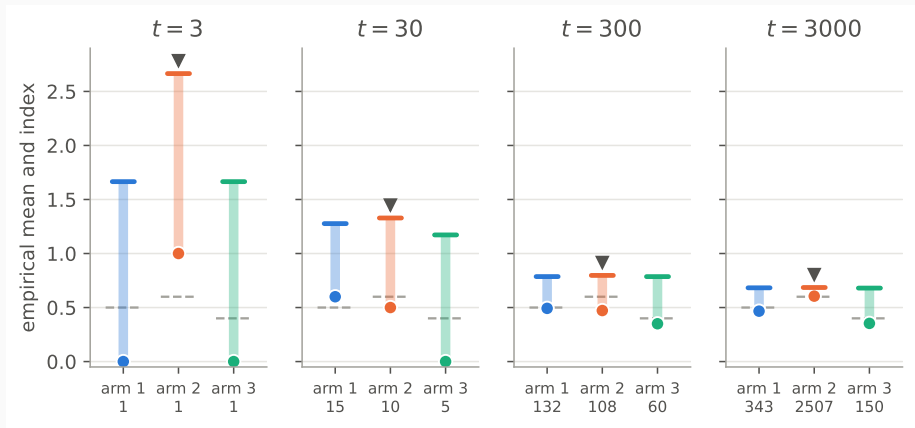
Three Bernoulli arms with means (0.5, 0.6, 0.4) (dashed), after  $t$  rounds. Dots: empirical means; bars: indices  $U_a(t)$ ; ▼: arm played at the next round; below each arm: number of pulls.

# One run of UCB1



Three Bernoulli arms with means (0.5, 0.6, 0.4) (dashed), after  $t$  rounds. Dots: empirical means; bars: indices  $U_a(t)$ ; ▼: arm played at the next round; below each arm: number of pulls.

# One run of UCB1

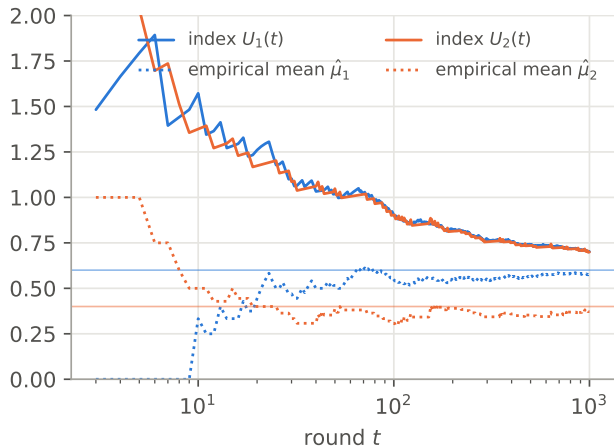


Three Bernoulli arms with means (0.5, 0.6, 0.4) (dashed), after  $t$  rounds. Dots: empirical means; bars: indices  $U_a(t)$ ;  $\blacktriangledown$ : arm played at the next round; below each arm: number of pulls.

# Indices along a run

UCB1,  $\mu = (0.6, 0.4)$ , one run of 1000 rounds:

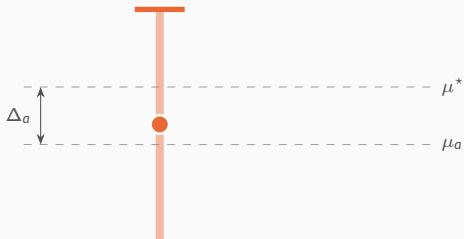
- the index of an arm drops when it is pulled and grows otherwise
- the two indices stay close to each other, above  $\mu^*$
- the bad arm is pulled each time its index catches up: 129 pulls out of 1000, less and less often



# Regret

---

# Pulls of a suboptimal arm

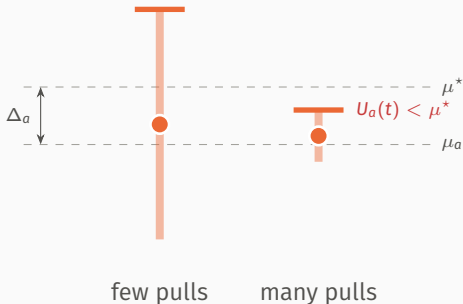


few pulls

When the confidence intervals hold:

- $U_{a^*}(t) \geq \mu^*$  for the best arm
- $U_a(t) \leq \mu_a + 2 \times \text{bonus of arm } a$

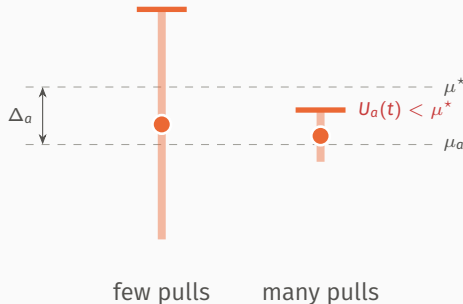
# Pulls of a suboptimal arm



When the confidence intervals hold:

- $U_{a^*}(t) \geq \mu^*$  for the best arm
- $U_a(t) \leq \mu_a + 2 \times \text{bonus of arm } a$
- once its bonus is below  $\Delta_a/2$ , arm  $a$  is not chosen anymore

# Pulls of a suboptimal arm



When the confidence intervals hold:

- $U_{a^*}(t) \geq \mu^*$  for the best arm
- $U_a(t) \leq \mu_a + 2 \times \text{bonus of arm } a$
- once its bonus is below  $\Delta_a/2$ , arm  $a$  is not chosen anymore

## Tutorial 2, exercise 2

$$N_a(T) \leq \frac{8 \log T}{\Delta_a^2} + 1$$

The cost  $\log(1/\delta)/\Delta^2$  of separating two arms (session 1) is paid without knowing  $\Delta_a$ .

## Theorem (Auer, Cesa-Bianchi and Fischer, 2002)

For rewards in  $[0, 1]$  and any horizon  $T$ , UCB1 satisfies

$$R_T \leq \sum_{a: \Delta_a > 0} \frac{8 \log T}{\Delta_a} + \left(1 + \frac{\pi^2}{3}\right) \sum_{a=1}^K \Delta_a .$$

- First term: regret decomposition and the bound on  $N_a(T)$ . Second term: rounds at which a confidence interval fails
- The regret is **logarithmic** in  $T$ , on any instance, with no tuning
- For  $\mu = (0.6, 0.4)$  and  $T = 5000$ , the bound is about 340

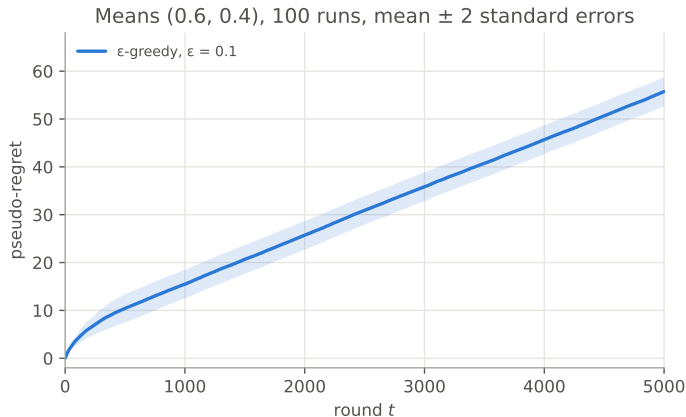
### Theorem (Lai and Robbins, 1985)

For Bernoulli arms, any strategy whose regret is  $o(T^p)$  for any  $p > 0$  on any instance satisfies

$$\liminf_{T \rightarrow \infty} \frac{R_T}{\log T} \geq \sum_{a: \Delta_a > 0} \frac{\Delta_a}{\text{kl}(\mu_a, \mu^*)} , \quad \text{kl}(p, q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q} .$$

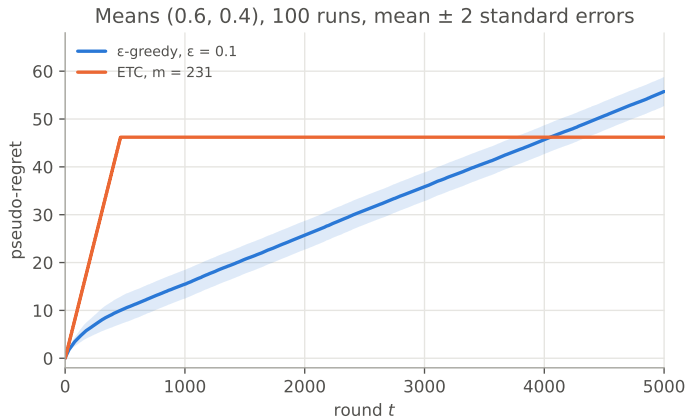
- The rate  $\log T$  cannot be improved
- For  $\mu = (0.6, 0.4)$ :  $\text{kl}(0.4, 0.6) \approx 0.081$ , so the lower bound is about  $2.5 \log T$ , against  $40 \log T$  in the bound of UCB1
- The constant of UCB1 is not optimal: refined indices (KL-UCB) and Thompson sampling reach the lower bound

## Comparison with the strategies of session 1



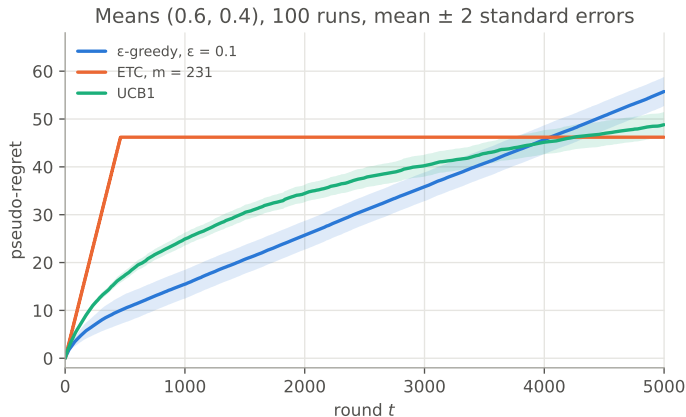
ETC uses  $m^*$ , computed from  $\Delta$  and  $T$ . UCB1 uses neither, and its measured regret ( $\approx 49$ ) is far below the bound of the theorem ( $\approx 340$ ).

## Comparison with the strategies of session 1



ETC uses  $m^*$ , computed from  $\Delta$  and  $T$ . UCB1 uses neither, and its measured regret ( $\approx 49$ ) is far below the bound of the theorem ( $\approx 340$ ).

## Comparison with the strategies of session 1

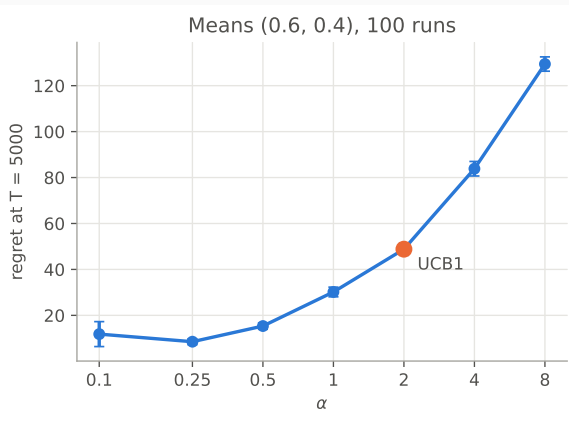


ETC uses  $m^*$ , computed from  $\Delta$  and  $T$ . UCB1 uses neither, and its measured regret ( $\approx 49$ ) is far below the bound of the theorem ( $\approx 340$ ).

# The parameter $\alpha$

UCB( $\alpha$ ): bonus  $\sqrt{\alpha \log(t)/N_a(t-1)}$ .

- Large  $\alpha$ : wide intervals, too much exploration
- Small  $\alpha$ : close to greedy. The mean regret is small on this instance, but a few runs stay on the wrong arm for a long time
- The theorem is stated for  $\alpha = 2$ . Smaller values often work better in practice (lab)



# Summary

| Strategy                                   | Needs to know    | Regret                       |
|--------------------------------------------|------------------|------------------------------|
| Greedy                                     | nothing          | linear (locking)             |
| $\varepsilon$ -greedy, fixed $\varepsilon$ | nothing          | linear                       |
| ETC, $m \approx T^{2/3}$                   | $T$              | $O(T^{2/3})$                 |
| ETC, $m = m^*$                             | $T$ and $\Delta$ | $O(\log(T)/\Delta)$          |
| UCB1                                       | nothing          | $O(\sum_a \log(T)/\Delta_a)$ |

## Tutorial 2

- UCB1 without noise
- Bound on  $N_a(T)$
- UCB1 by hand on three arms

## Labs 1 and 2

- Greedy,  $\varepsilon$ -greedy, ETC
- UCB, growth in  $\log t$ , effect of  $\alpha$  and of the gap

## Reading

- Lattimore and Szepesvári, *Bandit Algorithms*, chapters 7 and 16